

React Day 1

1. What is react ?

- React is javascript Library
- use to make frontend web application
- with the help of react we make the SPA(Single Page Application)

2. What is Library and Framework ?

Library:

- you are the boss
- the library is the collection of ready made code(function or tools) that we can pick and use when we want
- we control flow means we decide when and how to use library
- example : when we use react , we decide when to call components and what should happen next .

Framework:

- The framework is the boss
- The framework gives you the complete structure and tells where to write this code
- The framework control the flow, you just fill in the pieces
- example express , we just plug the logic in the structure that they have provided to us, express js takes care of the, listening to https server, sending response, managing the middleware

2. Why React js ?

- fast because of virtual DOM
- Large community and more demand in market

DOM(Document object Model) ?

- Its basically tree Structure thats basically represent the entire webpage.
- in simple words we can say that the DOM is the way our browser to understand the basic structure and organization the element of our webpage
- like in html the root node will be the html then the head and body are the childrens of the head node html

Virtual DOM

when we work with the **real DOM**, if we make a change to even **one element**, the browser often has to update or re-render **the entire DOM tree** or a large part of it. This process can be **slow and inefficient, especially for big web pages or complex applications**.

However, when we use the **Virtual DOM** (like in React), the process is much smarter and faster.

How it works:

- The **Virtual DOM** is a lightweight **copy** of the real DOM.
- When we make changes, React first **updates the Virtual DOM**, not the real one.
- It then **compares the updated Virtual DOM with the old Virtual DOM** to find out exactly what has changed.
(This process is called "**diffing**.")
- Finally, React updates **only the parts of the real DOM that actually changed**, instead of refreshing the whole tree.

Benefits

- Faster updates.
- Smoother user experience.
- Better performance, especially for large web apps.

More About React ?

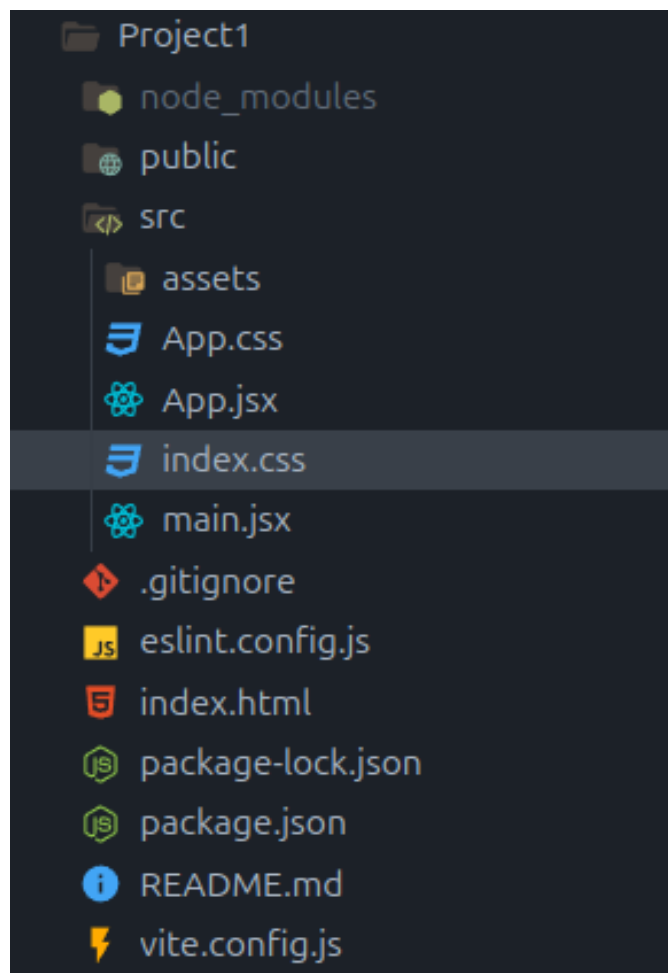
- Developed by facebook
- current version 19

Why we use Vite tool for react setup ?

- the vite is the fast development server and build tool
- simple configuration
- efficient production build

The first command we need is → `npm create vite@latest`

`then npm cd <project_name>, npm i, npm run dev`



It creates the folder structure like that

Code Flow for react

- Index.html
 - Main.jsx basically component render karte tya madhe app.jsx aste
 - then component App.jsx
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JSX: JavaScript XML

TSX: TypeScript XML
